

Fig. 3: GAN generator

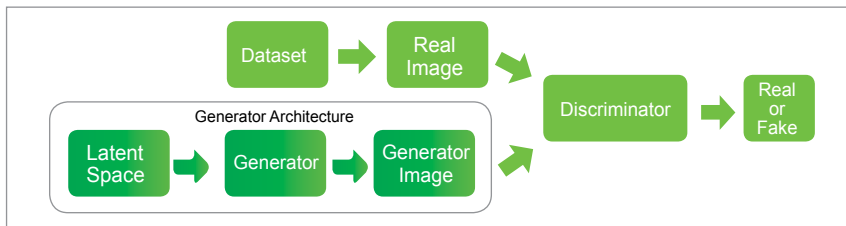


Fig. 4: GAN discriminator

with a tonic note, leaps resolved—in order to avoid awkward intervals, leaps are taken in opposite directions and leaping two times in the same direction is negatively rewarded, and composition with unique max-note and min-note—notes in motif which are succession of notes representing a short musical idea. These metrics form a music theory rule. The degree of improvement of these metrics is determined by the reward given to a particular behaviour.

The choice of metrics and weights determines the shape of the music created. The most recent model of magenta has used GAN and transformers to generate music with improved long-term structure.

Challenges in music generation

The greatest challenge in generation of music is to be able to encode various musical features. Once that is accomplished, the generative music is supposed to follow the broader structure, dynamics, and rules of music. Musical dimensions such as timing and pitch have relative rather than absolute significance when it comes to how notes are placed in these dimensions.

Features such as dynamics (that tells the volume of the sound from the instrument) and timbre (that differentiates between notes having the same pitch and loudness) are difficult to encode. There are other features such as duration, rest, timing and pitch that

also are challenging to represent as extracted features.

What the future could hold

Various GAN models such as MidiNet, SSMGAN, C-RNN-GAN, JazzGAN, MuseGAN, Conditional LSTM GAN, and others have been attempted for composing melody. Other generative models like the VAE, flow based models, autoregressive models, transformers, RBMs, HMMs, and many others have been used in research in this area.

The future direction of melody generation is moving towards bringing in more musical diversity and structure-handling capacity. Better interpretability and human control are aimed for. Standardised test datasets and evaluation metrics, cross modal generation, composition style transfer, lyric-free singing, and interactive music generation are some future research directions in this area. **EFY**

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